

COMPREHENSIVE OPERATIONAL MANUAL · NATIONAL COMPETITION

MIT LiftLab 2026

From Corner Stores to Classrooms

From diagnosis to implementation: a complete manual for an educational program to transform consumption in Mexico's nanostores — curriculum, instruments, pilot design, and evaluation.

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How to Read This Manual

Parts I and II establish the problem and its foundation. Part III is the operational core: the how, session by session, with every instrument. Part IV explains how it is tested and measured. An evaluator in a hurry can go straight to Part III; a rigorous one will find in Part IV the validation design.

Executive Summary

It is not ignorance or lack of will. It is a structural barrier — perceived price and availability — reinforced by a supply chain that barely offers healthy products. People already know and already want to do better; what is missing are the conditions and habits to act.

This manual responds to the MIT LiftLab question and, above all, proposes an operational and verifiable solution. The diagnosis (Part I) identified an intention-action gap of 39.1 percentage points: 85% of shoppers know that unhealthy food harms their health, and 63% say they would buy healthy options, but only 24% actually do — held back by price perception and an environment that nudges toward unhealthy choices.

The solution (Part III) is a one-school-year educational program that reinforces Physical Education and the "Healthy Living" strand of the NEM curriculum in secondary school, with four fully developed components: a 32-session curriculum that trains decision-making capacity, a field audit of the students' own environment, certified teacher training, and a low-time family bridge. Part V details the quasi-experimental pilot, indicators, budget (≈MXN\$47 per student), and risk management.

What this manual delivers:

Deliverable	Description
Diagnosis	4 analytical acts with real data and a Monte Carlo impact model
Curriculum	32 sessions in 4 blocks, with detailed lesson plans and rubrics
Instruments	Audit form, logbook, label cards, family materials
Training	12-hour teacher program with facilitator guide and certification
Evaluation	Pilot design with control group, KPIs, and pre/post measurements
Viability	Itemized budget, risk matrix, and curricular alignment

PART I

Diagnosis

The problem in four acts: from raw data to root cause.

Epidemiological and Regulatory Context

The MIT LiftLab question does not arise in a vacuum. Mexico is experiencing one of the most severe obesity crises in the world, and the educational system has a mandate — rarely implemented — to respond. This section establishes the magnitude of the problem and the legal framework that enables the intervention.

1. The Magnitude of the Problem

According to the ENSANUT Continua 2020–2024, the combined prevalence of overweight and obesity reaches approximately 37% in children aged 5 to 11 and around 40% in adolescents aged 12 to 19. Childhood obesity is not a passing problem: longitudinal evidence (Singh et al., 2008) shows that a child with obesity has a substantially higher probability of becoming an adult with obesity, with the associated risk of type 2 diabetes, hypertension, and cardiovascular disease.

Indicator	Group	Approx. Prevalence
Overweight + obesity	Children 5–11 years	~37%
Overweight + obesity	Adolescents 12–19	~40%
Persistence into adulthood	Children with obesity	Elevated risk (Singh 2008)

2. The School Food Environment

The nanostores surrounding schools are the immediate point of sale for millions of students. Our diagnosis showed that this environment is dominated by sugary drinks and snacks, that healthy options are scarce and perceived as more expensive, and that the supply chain itself under-supplies healthy products. Adolescents make decisions in that environment — not in an ideal one.

3. The Nueva Escuela Mexicana Mandate

The Plan de Estudio 2022 established "Healthy Living" as a transversal articulating strand of basic education and maintains Physical Education as a curricular space. However, this strand is rarely translated into concrete, measurable practices. There is, then, a gap between the normative mandate and its implementation in the classroom — precisely the space this proposal seeks to fill, without the need for legislative reform.

The Opportunity: The problem is large and persistent; information alone has not solved it; and there is a current but under-implemented curricular mandate. The intervention sits exactly at that intersection: high need, evidence that education can help, and an institutional vehicle already available.

The Diagnosis in Four Acts

Before proposing a solution, we had to understand the problem without shortcuts. We followed a four-act path using a national MIT LiftLab dataset (3,040 records) and our own survey (104 responses). We documented even the model that failed, because that failure led us to the right answer.

Act 1 — The Regression That Did Not Work

We began by trying to predict shopper spending with a linear regression on demographics and budget. The result was $R^2 = 0.032$: the model explains practically nothing. Far from being a failure, this was the first finding: behavior is not a linear function of who the person is. The problem is structural.

Act 2 — Machine Learning and the Complete System

We changed the question and analyzed the system. Four insights:

- Insight 1 — Unhealthy dominates. 60.8% buy at least one unhealthy product and 50.3% buy only unhealthy items; only 23.7% take something healthy.
- Insight 2 — The intention-action paradox. The core of the problem:

Indicator	%
Know that unhealthy food harms their health	85.3%
Say they would buy healthier	62.8%
Want more healthy options	64.6%
Perceive healthy as more expensive (barrier)	65.8%
Actually bought something healthy	23.7%

The gap between intention (62.8%) and action (23.7%) is 39.1 percentage points. That is the target of the entire intervention.

- Insight 3 — Even ML cannot predict purchase. A Random Forest achieved ROC-AUC ≈ 0.55 (near chance): healthy purchasing is not explained by individual traits, but by the environment.
- Insight 4 — Supply chain pushes unhealthy options. 66% of last-mile deliveries did not offer healthy products to the store. The bias is embedded before the counter.

Act 3 — Our Own Survey (Triangulation)

104 independent responses in San Luis Potosí confirmed the diagnosis: the most cited reason for not cooking was lack of time, and among the top three solutions most requested by the population itself was nutritional education. 80 of 94 people would change their habits with more time or resources.

Act 4 — Impact Simulation

Since we could not run a multi-year pilot before the competition, we simulated the impact of the intervention with a Monte Carlo model: a virtual cohort of 1,000 students, 5,000 runs per scenario, with effect sizes taken from published meta-analyses (Silveira 2013: -0.33 kg/m^2 for nutritional education, verified against the original article).

Scenario	Obesity at Year 5	vs. No Intervention
No intervention	31.9%	—
Conservative	29.8%	−2.1 pp
Central	26.2%	−5.7 pp
Optimistic	22.7%	−9.2 pp
Null robustness (Hodder)	32.0%	≈ 0

We deliberately included the null scenario: the evidence for combined interventions is mixed, and concealing it would be dishonest. The complete technical detail of the model is in the attached technical report.

PART II

Theoretical Framework

Why an educational program can close a gap that information alone could not.

Theoretical Framework: Why Education Can Close the Gap

The diagnosis posed a paradox: people know and want to change, but do not act. Why believe that education will change anything, if information is already available? The answer lies in distinguishing information from the competence to decide. Three frameworks from health psychology underpin the design of the intervention.

1. Health Belief Model (Rosenstock)

This model holds that health behavior depends not only on knowing that something is harmful, but on the perception of barriers and benefits. Our population perceives an enormous barrier — price — that outweighs their knowledge of the harm. The intervention acts directly on that perception: it teaches students to find affordable healthy options and to plan with a budget, thereby reducing the perceived barrier.

2. Social Cognitive Theory (Bandura)

This introduces self-efficacy: the belief that one is capable of executing a behavior. Knowing that soda is harmful is not enough if the adolescent does not feel capable of choosing differently at the corner store. This is why the program does not merely inform: it drills — through simulations, the 10-second pause, the 20-peso basket — building self-efficacy through repeated practice.

3. Self-Determination Theory (Deci and Ryan)

This distinguishes external motivation (obeying a rule) from internal motivation (deciding out of genuine conviction). Prohibiting generates fragile compliance; developing autonomy generates lasting change. The program never says "don't eat that": it provides tools for students to decide for themselves, fostering autonomy, competence, and belonging (by working as a team and with their family).

The intervention thesis: Information has already saturated the population and has not changed behavior. What is missing is not knowledge, but self-efficacy and autonomy to decide within an adverse environment. This is why the intervention is not another information campaign: it is training in the capacity to decide, aimed at those still forming their habits — adolescents.

4. The Generational Bridge

The diagnostic data are from adult consumption; the intervention targets adolescents. The logic is deliberate: in adults, habits are already consolidated and information did not move them; in adolescents, who will be the nanostore shoppers in 5–10 years, habits are still forming. The literature on obesity persistence (Singh et al., 2008) confirms that early intervention yields the greatest return. This is the point of maximum-efficiency leverage.

PART III

How the Models Were Built

Data engineering, regression, machine learning, and Monte Carlo simulation — with all parameters justified.

Data Preparation

Before any model comes a preparation stage that conditions everything else. Documenting it is part of methodological honesty: a result is only as good as the data feeding it.

Data Sources

Source	What it captures	Key structured variables	n
nanostore	Physical census of stores	store_size, location_type, street_type, coordinates	221
shopper	Shoppers	products_purchased, total_spent_numeric, healthy_price_perception, would_choose_healthier, daily_unhealthy_affects_health, age_group, monthly_food_budget	1,135
shopkeeper	Store owners	offers_healthy, top_unhealthy_products, supplier_help_needed	236
lastmile	Supply chain	offered_healthy, vehicle_type, rejected_products	1,448
Own survey	Habits and barriers (SLP)	reason for not cooking, solutions, willingness to change	104

Cleaning Decisions (and Why)

- BOM removal. CSVs carry a byte order marker (\uffff) at the start of the header; it is removed so column names match in the code.
- Numeric coercion. Columns such as monthly_food_budget mix text and numbers; they are converted with `pd.to_numeric(errors='coerce')` and nulls are imputed with the median (robust to outliers).
- Missing categoricals. Blanks are labeled "Missing" instead of being discarded: the absence of a response is information.
- Structured priority. Free text (e.g., healthy_products_bought) contains field noise (responses like "d" or "Fgghh"); structured multiple-choice columns are prioritized.

Model 1 — Linear Regression

Objective: Predict the shopper's spending per visit (total_spent_numeric) from observable variables.

Variables and Encoding

Type	Variables	Transformation
Categorical (X)	age_group, sex, purchase_frequency	One-Hot Encoding
Numerical (X)	household_size, monthly_food_budget, purchase_freq_monthly	Median imputation
Target (y)	total_spent_numeric	None

One-Hot Encoding converts each category into a binary column (0/1) so the model does not assume a non-existent numerical order. A ColumnTransformer applies this encoding to categoricals and passes through numerics, all chained in a scikit-learn Pipeline.

Cross-Validation and Metric

To avoid measuring fit on the same training data, 5-fold cross-validation is used: the data is split into five blocks; the model trains on four and evaluates on the fifth, rotating. The metric is the coefficient of determination R^2 :

$$R^2 = 1 - (SS_{\text{residual}} / SS_{\text{total}})$$

where SS_{residual} is the sum of squared errors of the model and SS_{total} is the total variance. $R^2 = 1$ is perfect prediction; $R^2 = 0$ is equivalent to always predicting the average.

Result

Average $R^2 = 0.032$ (by fold: 0.031, 0.038, -0.003 , 0.056, 0.040). The model explains ~3% of variation: noise. Correlations of spending with numerical variables are ~0.

Why this "failure" is a result: An R^2 near zero proves that behavior is not a linear function of who the person is. That is the evidence that justifies abandoning the individual approach and analyzing the system (Model 2).

Model 2 — Machine Learning

We changed the question to: does the shopper buy healthy products? To define this reliably, we classified the structured column `products_purchased` with the following taxonomy:

Class	Categories
Unhealthy	Sugary drinks, Salty snacks, Bakery items, Cigarettes
Healthy	Fruits and vegetables, Bottled water
Neutral	Bread/Tortillas, Dairy

Technical decision documented: We discarded the free-text column `healthy_products_bought` as the target because its noise inflated the healthy purchase rate to 80%. With the structured column, the real rate is 23.7% — consistent with the rest of the evidence. Documenting this correction is part of rigor.

The Random Forest Model

To see whether healthy purchasing is predicted by individual traits, we trained a Random Forest: an ensemble of many decision trees, each trained on a different sample of data and variables; the final prediction is the average vote, which reduces the overfitting of a single tree.

Hyperparameter	Value	Justification
<code>n_estimators</code>	400	Enough trees to stabilize the vote
<code>class_weight</code>	balanced	Compensates for only 23.7% being healthy buyers (minority class)
<code>random_state</code>	42	Reproducibility
Validation	5-fold CV	Honest out-of-sample evaluation
Metric	ROC-AUC	Robust to class imbalance

ROC-AUC measures how well the model separates the two classes: 1.0 = perfect, 0.5 = chance. Result: 0.545 — barely above chance. That even a Random Forest cannot predict the purchase is, in itself, a finding: behavior is not explained by who the shopper is, but by the environment.

The Four Insights (Quantitative Summary)

Insight	Finding	Figure	
1. Unhealthy dominates	Buy ≥ 1 unhealthy / only unhealthy	60.8% / 50.3%	
2. Paradox	Know / would buy / see it as expensive / buy healthy	85% / 63% / 66% / 24%	
3. ML does not predict	ROC-AUC of Random Forest	0.545	
4. Supply bias	Deliveries without healthy offering	66%	

The intention-action gap (63% – 24%) is 39.1 percentage points: the behavioral objective of the entire intervention.

Model 3 — Monte Carlo Simulation

Why Monte Carlo: There are no longitudinal data from this intervention in Mexico, so a predictive model cannot be trained. Monte Carlo simulation is the honest option: repeat a random experiment thousands of times with parameters from published meta-analyses, to obtain a distribution of results with their uncertainty — not a single number.

Step 1 — Cohort Initialization

A virtual cohort of 1,000 students is created with initial BMI drawn from a Normal distribution calibrated with population data:

$BMI_initial \sim \text{Normal}(\text{mean} = 21.8, \text{sd} = 4.1)$

Step 2 — Source of Uncertainty (sigma)

Instead of a fixed effect, in each run the effect size is drawn from a Normal centered on the meta-analysis value. The width (sigma) is derived from the 95% confidence interval: since a 95% CI spans ~1.96 deviations on each side (total range 3.92):

$\text{sigma} = (\text{upper_limit} - \text{lower_limit}) / 3.92$

Thus, the uncertainty of the model is exactly that measured by the original studies — it is not invented.

Step 3 — Effect Parameters and Sources

Component	BMI Effect	95% CI	Source (verified)
Nutritional education	-0.33	(-0.55, -0.11)	Silveira 2013, PMID 23370048
Physical activity + nutrition	-0.17	(-0.29, -0.06)	Cochrane (school review)
Adolescent health (BMI-z)	-0.06	(-0.10, -0.03)	Brown 2020
Null result (robustness)	0.00	(-0.05, 0.04)	Hodder 2021

Honesty about parameters: The effect of nutritional education from Silveira (-0.33) was verified against the original article. The evidence for combined interventions is mixed — some large meta-analyses find modest or null long-term effects — which is why we deliberately included the null scenario (Hodder 2021). We do not present only favorable scenarios.

Step 4 — Year-by-Year Recurrence

For each student and each year, BMI evolves by adding natural growth and subtracting the effect of the intervention (adjusted for adherence and dose):

$BMI(t+1) = BMI(t) + \text{annual_growth} + \text{effect} \times \text{adherence} \times \text{dose}$

Natural growth is drawn each year from Normal(0.45, 0.25); default adherence is 0.80 (80% compliance) and dose is 1.0 (reference hour per week). The effect is negative, thus reducing BMI.

Step 5 — The Five Scenarios

Scenario	Effect Used	Logic
Baseline	0.0	No intervention (control)
Central	Normal(-0.17, sigma)	Typical combined intervention
Optimistic	Normal(-0.29, sigma/2)	Lower bound of CI = greatest reduction
Conservative	Normal(-0.06, sigma/2)	Upper bound of CI = smallest reduction
Null robustness	Normal(0.00, sigma)	Null long-term effect (Hodder)

Step 6 — Algorithm Pseudocode

```
for each scenario in {baseline, central, optimistic, conservative, null}:  for run
= 1..5000:      cohort <- 1000 values ~ Normal(21.8, 4.1)      # initial BMI      effect
<- draw from Normal(mu_scenario, sigma)      for year = 1..5:      growth <-
Normal(0.45, 0.25) per student      cohort <- cohort + growth +
```

effect*adherence*dose record % overweight (BMI>=22.5) and % obesity (BMI>=26.0)
average 5000 runs -> distribution of results

Step 7 — Why $1,000 \times 5,000$

One thousand students stabilizes prevalences within each run; five thousand runs makes averages converge across runs (law of large numbers). Each scenario evaluates 5 million trajectories. With fixed seed (42), results are perfectly reproducible.

Results and Sensitivity

Scenario	Obesity at Year 5	vs. Baseline
No intervention	31.9%	—
Conservative	29.8%	−2.1 pp
Central	26.2%	−5.7 pp
Optimistic	22.7%	−9.2 pp
Null robustness	32.0%	≈0

The sensitivity analysis varies adherence (50%–100%) and dose to show how much the results depend on those assumptions: this shields the model from the criticism of having chosen convenient numbers.

Technical Appendix — Parameter File, Justified Field by Field

All model assumptions live in a single configuration file (`literature_parameters.json`), separated from the code so they are auditable and modifiable without touching the logic. Each value has a source. This is its commented structure:

```
# literature_parameters.json (structure) intervention_effects:  nutrition_education: -0.33
CI(-0.55,-0.11)  Silveira 2013  physical_activity_plus_nutrition: -0.17  CI(-0.29,-0.06)
Cochrane  adolescent_health (BMI-z): -0.06  CI(-0.10,-0.03)  Brown 2020  long_term_null:
0.00  CI(-0.05, 0.04)  Hodder 2021  behavioral_targets:  baseline_healthy_purchase_rate:
0.237  # Act 2 (real data)  baseline_intention_rate: 0.628  # Act 2 (real data)
intention_action_gap: 0.391  # target gap  baseline_population:  mean_bmi_age_12: 21.8
sd_bmi: 4.1  # ENSANUT-derived  overweight_threshold: 22.5  obesity_threshold: 26.0  # WHO
annual_bmi_growth: 0.45  sd: 0.25  simulation_settings:  cohort_size: 1000
n_runs_per_scenario: 5000  horizons_years: [1,3,5]  random_seed: 42  default_adherence: 0.80
default_dose: 1.0  sensitivity_sweeps:  adherence_rates: [0.5,0.6,0.7,0.8,0.9,1.0]
dose_hours: [0.5,1.0,1.5,2.0,3.0]
```

The separation of parameters/code allows an evaluator to question an assumption, change a single number, and re-run the simulation to see the effect — without rewriting anything. It is reproducibility and transparency by design.

Technical Stack

Layer	Tool	Use
Language	Python 3.10+	All analysis
Data	pandas, numpy	Loading, cleaning, transformation
Modeling	scikit-learn, statsmodels	Regression, Random Forest, metrics
Statistics	scipy	Distributions and sampling
Visualization	matplotlib, seaborn	Charts
Report / web	reportlab, Chart.js	PDF and dashboard

PART IV

Operational Manual of the Intervention

The heart of the project: the how, component by component, session by session.

Component 1 — Expanded Physical Education: the "Choose to Live" Program

This is the heart of the intervention. The additional Physical Education hour is not devoted to more physical exercise, but to what we call decision literacy: the concrete capacity of an adolescent to read a nutrition label, compare prices, understand what a sugary drink does to their body, and plan what to buy at the corner store before they have money in hand. We attack directly the intention-action gap of 39.1 percentage points documented in the diagnosis.

1.1 — General Objective and Competencies

General objective: By the end of the school year, secondary school students will be able to make informed and autonomous consumption decisions in their real environment (the nanostore), applying criteria of health, budget, and self-regulation.

The program develops four competencies, aligned with the NEM graduate profile:

- Competency 1 — Applied nutritional literacy: interpret nutrition labels, warning seals, and food advertising to distinguish options based on their health impact.
- Competency 2 — Consumption self-regulation: recognize triggers of impulse buying and apply strategies to resist them.
- Competency 3 — Critical reading of the environment: analyze the supply, prices, and advertising of their own community.
- Competency 4 — Planning and economic agency: design a feasible healthy consumption plan with a real, limited budget.

1.2 — Program Structure (Scope and Sequence)

The program is delivered in one 50-minute weekly session over 32 weeks of the school year, organized into four progressive thematic blocks. The progression goes from the personal (my body) to the critical (my environment) and ends in agency (my plan, my family).

Block	Weeks	Title	Guiding Question
1	1–8	My Body in Motion	What does what I eat do inside me?
2	9–16	Reading to Choose	How do I know what I'm actually buying?
3	17–24	The Corner Store and Me	What does my environment offer — and hide — from me?
4	25–32	My Plan, My Family	How do I decide better with what I have?

Pedagogical principle: Each block closes with an observable product (an analysis, a form, a presentation, a plan). We do not evaluate what students can say, but what they can do — because the diagnosis showed that knowing is not enough to act.

1.3 — Block 1: "My Body in Motion" (Weeks 1–8)

Block purpose: to establish, from experience and not from scolding, what happens in the body with what is consumed and why movement matters. Sets the groundwork for self-regulation.

Session 1 — What Do I Decide Without Realizing? (50 min)

- Objective: Students become aware of how many consumption decisions they make automatically each day.
- Expected learning: Identifies their own consumption habits at the corner store and emotional triggers.
- Opening (10 min): "Map of my week" activity — each student draws what they bought at the corner store over the last 7 days.
- Development (25 min): In teams, classify purchases as healthy / unhealthy / neutral. Discussion: why did you buy that? Introduction of "impulse buying."
- Closing (15 min): Each student writes in their logbook a personal observation goal (not a change goal) for the week.
- Assessment: Checklist — student completes their map and formulates an observation goal (formative).

Session 3 — The Soda Journey (50 min)

- Objective: Understand what happens in the body when consuming sugary drinks — without alarmism, with evidence.
- Expected learning: Explains the effect of added sugar and relates consumption to energy balance.
- Opening (10 min): Visual challenge — guess how many teaspoons of sugar each drink contains; revealed by measuring real sugar with spoons.
- Development (25 min): Guided explanation of energy balance (in/out) with everyday physical activity examples. Calculation: how much walking does it take to "burn off" a soda?
- Closing (15 min): Logbook: student records yesterday's drinks and estimates their sugar.
- Assessment: Product — estimation of sugar from their own consumption, with justification.

Session 6 — My Impulse Thermometer (50 min)

- Objective: Recognize personal triggers of impulsive buying and rehearse a pause strategy.
- Expected learning: Applies the "10-second pause" technique when faced with a simulated trigger (self-regulation).
- Opening (10 min): Brainstorm — when do I buy without thinking? (hunger, craving, peer pressure, advertising).
- Development (25 min): Role play — faced with situation cards, students practice the "10-second pause" and a self-control question ("do I need it or do I want it?").
- Closing (15 min): Weekly commitment — use the pause at least 3 times and log it in the notebook.
- Assessment: Performance rubric in role play + logbook record.

Block 1 closing product: personal consumption logbook with at least 4 entries and a written half-page reflection on personal impulse triggers.

1.4 — Block 2: "Reading to Choose" (Weeks 9–16)

Block purpose: equip students with the central technical tool — how to read a label and a warning seal — so that decisions no longer depend on brand and flavor. This addresses practical misinformation at its root.

Session 9 — Decoding the Label (50 min)

- Objective: Read and interpret the nutrition facts table and warning seals of Mexican labeling.
- Expected learning: Identifies sugars, sodium, and calories per serving and distinguishes "serving" from "package."
- Opening (10 min): Show two products that "seem" healthy by their packaging but have warning seals; contrast.
- Development (30 min): Hands-on workshop — in pairs, fill in the "reading form" for 3 real products (sugars, sodium, seals, servings per package). Teacher models the first one.
- Closing (10 min): Plenary — which product was most "deceptive"? Why does serving size matter?
- Assessment: Product — 3 correctly completed reading forms (accuracy rubric).

Session 12 — Advertising Is Not Your Friend (50 min)

- Objective: Critically analyze marketing strategies aimed at young people.
- Expected learning: Detects persuasive techniques (characters, gifts, "natural," size) in real ads.
- Opening (10 min): View 2–3 ads for unhealthy products aimed at youth.
- Development (30 min): In teams, identify techniques used and redesign the ad to tell "the truth" about the product (educational parody).
- Closing (10 min): Gallery — "honest" ads are exhibited.
- Assessment: Product — redesigned ad + list of detected techniques.

Block 2 closing product: "Quick Guide to Staying Smart" — a one-page infographic, made by teams, summarizing how to read a label and detect misleading advertising.

1.5 — Block 3: "The Corner Store and Me" (Weeks 17–24)

Block purpose: bring the analysis into the student's real environment. This block is linked to Component 2 (store audit): sessions prepare, accompany, and process the fieldwork. This is where learning becomes research.

Session 17 — Preparing the Expedition (50 min)

- Objective: Understand the audit protocol and the use of the field form.
- Expected learning: Applies the observation instrument in a simulated exercise in the classroom.
- Opening (10 min): Framing — "today you become researchers of your own neighborhood." Safety and respect rules for the storekeeper.
- Development (30 min): Simulation — with a "corner store" set up in the classroom, teams practice filling in the form (counting products, recording healthy vs. unhealthy prices).
- Closing (10 min): Assignment of zones and teams; signing of the conduct letter.
- Assessment: Checklist — mastery of the instrument in the simulation.

Session 20 — What We Found (50 min)

- Objective: Systematize and analyze the data collected in the field audit.
- Expected learning: Organizes data in a table, calculates proportions, and detects patterns.
- Opening (10 min): Each team shares one data point that surprised them.
- Development (30 min): Data entry into the shared template — % of unhealthy products, average price difference healthy vs. unhealthy, location of products in the store.
- Closing (10 min): Teams formulate a hypothesis: "why do they think the store arranges its products that way?"
- Assessment: Product — team data table with at least 3 correct calculations.

Block 3 closing product: "Neighborhood X-Ray" — a team poster with the audit findings (product proportion, prices, observations), presented to the class.

1.6 — Block 4: "My Plan, My Family" (Weeks 25–32)

Block purpose: convert everything learned into concrete agency. The student designs a real consumption plan with a real budget, and the bridge with the family is built (Component 4).

Session 26 — 20 Pesos, Good Decisions (50 min)

- Objective: Design the best possible purchase at the corner store with a real, limited budget.
- Expected learning: Optimizes a consumption decision under a budget constraint.
- Opening (10 min): Challenge — "you have 20 pesos for your entire week's recess, what do you buy?" using the real prices students themselves collected.
- Development (30 min): Each student builds their "basket" and justifies it; team comparison of who achieved the healthiest option without going over budget.
- Closing (10 min): Reflection — was it possible? What made it difficult? (Here the price barrier emerges, validated with data.)
- Assessment: Product — justified "20-peso basket" (health + budget criterion rubric).

Session 30 — My Change Project (50 min)

- Objective: Formulate a measurable and realistic personal commitment to change.
- Expected learning: Defines a SMART consumption goal and a tracking indicator.
- Opening (10 min): Review of the journey — from "I didn't realize" to "I know how to decide."
- Development (30 min): Each student drafts a SMART goal (specific, measurable, achievable) and how they will measure it over a month.
- Closing (10 min): Voluntary public pledge — whoever wants to shares their goal with the group.
- Assessment: Final product — personal change project with SMART goal and indicator.

Final program product: student portfolio gathering logbook, label forms, audit poster, and personal change project. It is the comprehensive evidence of the competency developed.

1.7 — Component Assessment System

Assessment is formative and evidence-based, not by exam. It consists of observable weighted products:

Evidence	Block	Weight	Main Criterion
Consumption logbook	1	15%	Consistency and honesty of recording
Label forms	2	20%	Accuracy in nutritional reading
Anti-deception infographic	2	10%	Clarity and precision
Audit poster	3	25%	Data rigor and analysis
20-peso basket	4	10%	Health-budget balance
Change project	4	20%	SMART goal and feasibility

1.8 — Generic Performance Rubric (4 Levels)

Level	Descriptor
Strategic (4)	Applies the criterion autonomously, justifies it with evidence, and transfers it to new situations (e.g., advises a family member).
Autonomous (3)	Applies the criterion correctly without help in the practiced situation.
In progress (2)	Applies the criterion with support or with minor errors.
Initial (1)	Recognizes the criterion but does not yet apply it consistently.

1.9 — Complete 32-Session Calendar

Week-by-week map of the program. Sessions with detailed plans in this manual are marked with ★; the others follow the same structure (opening–development–closing) on the indicated topic.

Wk	Session	Wk	Session
1	★ What Do I Decide Without Realizing?	17	★ Preparing the Expedition
2	My Body Needs Energy	18	Field Trip 1
3	★ The Soda Journey	19	Field Trip 2
4	Salt, Fat, and Craving	20	★ What We Found
5	Real Hunger vs. Craving	21	Field Trip 3
6	★ My Impulse Thermometer	22	Price Maps
7	Sleep and Food	23	Why Is It Arranged That Way?
8	Block 1 Close: My Logbook	24	Block 3 Close: X-Ray
9	★ Decoding the Label	25	Planning the Week
10	Sugars by Other Names	26	★ 20 Pesos, Good Decisions
11	Sodium: The Silent Enemy	27	Cooking Is Deciding
12	★ Advertising Is Not Your Friend	28	The Real Healthy Plate
13	Warning Seals in Depth	29	My Family and I
14	Serving vs. Package	30	★ My Change Project
15	Comparing Two Products	31	Presenting My Portfolio
16	Block 2 Close: Infographic	32	Closing: Group Pledge

1.10 — Student Instruments

Consumption Logbook (Block 1).

Used by students all week. Structure per day:

MY CONSUMPTION LOGBOOK Name: _____ Week: ____ Day | What I bought at the corner
store | \$ | Healthy? | Why did I buy it? -----|-----|-----|-----|-----
--|----- Mon | | | Y / N | Tue |
| | Y / N | Wed | | | Y / N | ... My most
frequent impulse trigger was: _____ I used the "10-second pause" ____ times this
week.

Label Reading Form (Block 2).

One per analyzed product:

LABEL FORM Product: _____ Servings per package: ____ Sugar per
serving: ____ g If I eat the ENTIRE package, total sugar: ____ g (= ____ teaspoons) Sodium per
serving: ____ mg Calories per serving: ____ Warning seals: [] Excess sugar [] Excess sodium
[] Excess calories [] Excess fat My verdict: [] Keep buying [] Moderate [] Switch to:

Component 2 — Critical Environment Module: "Store Detectives"

If the diagnosis showed that the problem is structural — the supply and prices of the environment — the intervention must teach students to read that environment. This component turns students into field researchers of their own neighborhood, replicating at a school scale the same methodology used by the MIT LiftLab. It is project-based learning, aligned with the NEM's Critical Thinking strand.

2.1 — Audit Protocol (Step by Step)

The audit is conducted over three biweekly outings during Block 3, in teams of 4–5 students, always accompanied by a responsible adult and at stores previously authorized within a safe radius of the school.

- Authorization and safety: The school contacts and obtains permission from participating store owners; the safe radius is defined and zones are assigned per team. Each outing includes an adult chaperone.
- Structured observation: The team fills in the field form (section 2.2) without interfering with store operations — counts products by category, records prices and location.
- Optional mini-interview with the storekeeper: Three short, respectful questions (What sells most? Do your suppliers offer healthy products? Do customers request them?).
- Data entry: Back in the classroom (session S20), data are transferred to the shared template.
- Analysis and product: The team calculates proportions and prices, and builds their "Neighborhood X-Ray" poster.

2.2 — The Instrument: Nanostore Audit Form (Annex A)

This is the exact instrument filled in by each team. It is designed to be quick (10–15 min per store) and to produce quantifiable data.

AUDIT FORM — STORE DETECTIVES Team: _____ Date: _____ Store (code): T-____ A. LOCATION AND SIZE ☐ Corner ☐ Mid-block Street: ☐ Main ☐ Secondary Approx. size: ☐ <20 m2 ☐ 20–50 ☐ 51–100 ☐ >100 B. PRODUCT COUNT (visible at counter and entrance) Sugary drinks Bottled water Chips / snacks Fruit / vegetables ... Sweet bread / pastry . Dairy Candy / sweets Cigarettes C. PRICES (record the lowest price for each) Soda 600 ml \$_____ Water 600 ml \$_____ Chips bag \$_____ Fruit (piece/bag) \$_____ D. STRATEGIC PLACEMENT (check all that apply) ☐ Unhealthy items are at eye level / children's level ☐ Sugary drink advertising is at the entrance ☐ Healthy items (if any) are at the back or hidden E. MINI-INTERVIEW WITH STOREKEEPER (optional, respectful) 1. What sells most? _____ 2. Do your suppliers offer healthy products? ☐ Yes ☐ No 3. Do customers ask for healthy options? ☐ Yes ☐ Sometimes ☐ No

2.3 — Analysis Template

With completed forms, the team calculates three indicators that replicate those of the national study, allowing comparison of "my neighborhood" with "Mexico":

Indicator	How It Is Calculated	National Reference
% unhealthy supply	Unhealthy ÷ total products counted × 100	Majority of basket
Price gap	Price of soda – price of water (and chips – fruit)	Healthy perceived as more expensive (66%)
Visibility index	No. of signals from section D checked (0–3)	Supply biased toward unhealthy

2.4 — Rubric for the "Neighborhood X-Ray" Product

Criterion	Strategic (4)	Autonomous (3)
Data rigor	≥3 stores, complete and verified data	2–3 stores, complete data
Analysis	Calculates 3 indicators and interprets them	Calculates 3 indicators
Connection	Compares with national data and draws conclusions	Mentions national data
Communication	Clear, visual, well-argued poster	Clear poster

Double benefit: This audit not only teaches — it generates real data that the school can accumulate year after year to monitor its own food environment. The school becomes a research partner, not just a recipient.

Component 3 — Teacher Training: "Training the Trainer"

No program survives without trained teachers. The most common error in interventions — and one of the causes of null effects documented in the literature (Hodder 2021) — is handing the teacher a task without the tools. This component resolves this with brief, practical, and certified training.

3.1 — Structure: 12 Hours (8 Asynchronous + 4 In-Person)

The format respects the teacher's time: theory is covered online at their own pace, and the in-person session is reserved for practice.

Module	Format	Hours	Content
M1. The Problem and the Evidence	Asynchronous	2	The nanostore diagnosis, the intention-action gap, why educate rather than prohibit.
M2. Decision Literacy	Asynchronous	2	Label reading and seals, energy balance, food marketing aimed at youth.
M3. Program Pedagogy	Asynchronous	2	How to facilitate each block, logbook management, evidence-based assessment.
M4. Safety and Ethics of Fieldwork	Asynchronous	2	Audit protocol, consent, treatment of the storekeeper, managing groups outside the classroom.
M5. Hands-on Workshop	In-person	4	Simulation of 3 key sessions, practice with instruments, questions, and local planning.

3.2 — Facilitator Guide (Workshop M5 Excerpt)

The in-person workshop follows a minute-by-minute script to ensure consistency across schools:

Block	Min	Facilitator Activity
Framing	20	Presents the why with data; connects with the teacher's own school experience.
Modeling	60	Demonstrates live session "The Soda Journey" (S3); teachers are "students."
Practice	90	In teams, teachers facilitate a session fragment and receive peer feedback.
Instruments	40	Tour through logbook, label form, and audit form; questions answered.
Planning	30	Each teacher adapts the calendar to their context and schedules field trips.

3.3 — Certification and Teacher Assessment

The teacher earns credit by: (a) online quizzes in M1–M4 ($\geq 80\%$ correct), and (b) practical facilitation in M5. They receive a certificate with curricular credit, manageable with the state educational authority as continuing education. The certificate is also the incentive that recognizes the additional workload without turning it into unpaid work.

Component 4 — Family Bridge: "The Table Also Educates"

The diagnosis was clear: the main barrier reported by families is lack of time. That is why the family bridge is not another homework assignment or a mandatory meeting — it is a brief monthly invitation, designed to fit into a working family's schedule. A guided conversation, an affordable recipe, and a reflection sheet. Nothing that takes more than 30 minutes.

4.1 — Calendar of Nine Monthly Activities

Month	Family Activity	Link to Classroom
Sep	"Map of our pantry": review together what is at home	Block 1 (awareness)
Oct	"The hidden sugar challenge": find seals on products at home	Block 1 / S3
Nov	"Reading a label together" with the student's guide	Block 2 / S9
Dec	"Grandma's recipe vs. the package": compare cost and healthiness	Block 2
Jan	"Detectives at home": student shares their neighborhood X-ray	Block 3
Feb	"Smart shopping": plan a weekly shop together with a budget	Block 4 / S26
Mar	"Our 20-peso snack": prepare the student's basket as a family	Block 4
Apr	"The week's plate": cook an affordable recipe from the recipe book	Block 4
May	"Our commitment": family adopts a goal from the student's project	Block 4 / S30

4.2 — Developed Example: November Activity

Each activity arrives at home on a single sheet (front: conversation and recipe; back: reflection). This is the complete November sheet:

THE TABLE ALSO EDUCATES — November "Reading a label together" (estimated time: 20 min)
GUIDED CONVERSATION (student leads — they learned how in class) 1. Take a product you buy often. Find the nutrition facts table. 2. Ask together: how many servings does the package have? How much sugar is in ONE serving? And if we eat the whole package? 3. Does it have warning seals? How many? 4. Discuss: knowing this, would we buy it the same way again? RECIPE OF THE MONTH: Oat water (yields 2 L) — 4 tablespoons of oats — 1 L of water — cinnamon to taste — sweeten with a little sugar or fruit Approx. cost: \$12 (vs. \$30 for soda) Blend, strain, serve cold. Compare: how much do we save vs. soda? REFLECTION (back of sheet, one line per person) What surprised me most was: _____ This week at home we are going to try: _____

Why it works: The student leads the conversation: they apply at home what they learned and become an agent of change, not a recipient. The family invests 20 minutes, not an afternoon. And the recipe demonstrates with real money that healthy can be cheaper — deactivating the price barrier in the kitchen itself.

4.3 — Recipe Book: "The Table Also Educates"

Each family bridge recipe is costed and compared against its unhealthy alternative, to demonstrate with real money — not with rhetoric — that eating better can cost less. Costs are approximate and for family portions.

- Oat water · yields 2 liters · cost ≈ \$12 vs. 2L Soda (\$30) — Ingredients: 4 tbsp oats, 1L water, cinnamon, a little sugar or fruit. Blend oats with water, strain, add cinnamon, sweeten lightly. Serve cold.
- Bean and cactus tacos · yields 4 servings · cost ≈ \$28 vs. Fast food (\$120) — Ingredients: cooked beans, cactus (nopales), tortillas, onion, salsa. Grill cactus with onion, heat beans, assemble tacos with salsa to taste.
- Fruit with lime and chili · yields 4 servings · cost ≈ \$25 vs. Chips (\$40) — Ingredients: jicama, cucumber, carrot, orange, lime, chili powder. Cut into sticks, squeeze lime, sprinkle chili. Ideal for recess.
- Oat quesadillas · yields 6 pieces · cost ≈ \$30 vs. Sweet bread package (\$45) — Ingredients: ground oats, water, salt, cheese, pan. Mix oats with water and salt to form dough, make tortillas, fill with cheese, grill.
- Egg and vegetable sandwich · yields 2 servings · cost ≈ \$18 vs. Store sandwich (\$35) — Ingredients: egg, whole-grain bread, tomato, lettuce, avocado. Cook egg, assemble sandwich with vegetables. Ready to go.
- Homemade popcorn · yields 4 servings · cost ≈ \$8 vs. Large chips (\$35) — Ingredients: popcorn corn, minimal oil, salt or lime. Pop corn in covered pot with little oil; season lightly. Healthy movie snack.

The argument that deactivates the price barrier: The diagnosis showed that 66% perceive healthy food as more expensive. Each recipe is a concrete refutation: oat water costs less than half the price of soda. The family does not receive a sermon — they receive proof they can touch and taste.

4.4 — Nine Family Activities, Developed

Month	Activity	Concrete Family Action (20 min)
Sep	Map of our pantry	Review together what is at home and classify as healthy / unhealthy; discuss what predominates and why.
Oct	The hidden sugar challenge	Find warning seals on 5 products at home; be surprised by which ones have them.
Nov	Reading a label together	Student teaches the family to read a nutrition label; prepare oat water.
Dec	Grandma's recipe vs. the package	Compare the cost and healthiness of a homemade dish against its packaged version.
Jan	Detectives at home	Student shares their "neighborhood X-ray"; family comments on which store they frequent.
Feb	Smart shopping	Plan a weekly shop together with a real budget, prioritizing healthy and affordable options.
Mar	Our 20-peso snack	Prepare as a family the "20-peso basket" the student designed in class.
Apr	The week's plate	Cook together a recipe from the recipe book; calculate how much was saved.
May	Our commitment	Family adopts a goal from the student's change project and sticks it on the fridge.

PART V

Implementation

How it is launched, tested, and measured.

Roadmap, Pilot, and Evaluation

1. Theory of Change

The program rests on an explicit causal chain: inputs that produce activities, which generate outputs, which lead to short- and medium-term results, and finally to long-term impact. Making it explicit allows each link to be evaluated separately.

INPUTS	ACTIVITIES	OUTPUTS	RESULTS	IMPACT
-----	-----	-----	-----	-----
Lower teachers	sessions	Trained students	32 PE intention-action	Literate obesity Materials
Field	Neighborhood gap	Family bridge	in the School	audits
Better	cohort partnership	Involved	consumption	at 5 yrs
Budget projected)	9 activities	families	decisions	(Monte Carlo

2. Phase Roadmap

Phase	Duration	Key Milestones
0. Preparation	2 months	Partnership with pilot school (Sec. 78), store authorization, material printing, teacher recruitment.
1. Training	1 month	12 h of teacher training, certification, calendar adjustment to local context.
2. Baseline	2 weeks	Pre-measurement: BMI, habits and intention survey, initial environment audit.
3. Implementation	1 school year	32 sessions, 3 audits, 9 family activities.
4. Post-measurement	2 weeks	Repeat measurements; comparison with control group.
5. Analysis and adjustment	1 month	Results evaluation, lessons learned, adjustments for scaling.

3. Pilot Design (Quasi-Experimental)

For results to be credible — not just anecdotal — the pilot is designed with a comparison group. The validation that Monte Carlo only projects, the pilot measures.

- Design: matched schools (or groups) — one receives the program (intervention), the other continues normally (control), chosen by socioeconomic similarity.
- Assignment: ideally random by group; if not possible, matching by observable characteristics.
- Pre and post measurements in both groups to estimate the net effect (difference-in-differences).
- Follow-up: minimum one school year, with a 12-month tracking measurement to capture attenuation (the risk warned by Hodder 2021).
- Ethical considerations: informed consent from parents, assent from minors, anonymized data, and program offer to the control group at the end (waiting list).

4. Evaluation Framework: Indicators (KPIs)

We measure three levels: what we did (process), what changed in behavior (intermediate outcome), and what changed in health (impact). The star indicator is the intention-action gap, because it is the variable the diagnosis identified as the cause.

Level	Indicator (KPI)	Instrument	Pilot Target
Process	% of sessions delivered	Teacher logbook	≥ 90%
Process	% of students with complete portfolio	Evidence review	≥ 80%
Process	% of families returning ≥3 sheets	School record	≥ 50%
Outcome	Reduction in intention-action gap	Pre/post survey	−10 pp

Outcome	% reading labels before buying	Pre/post survey	+20 pp
Outcome	Improvement in literacy test	Pre/post test	+25%
Impact	BMI change vs. control	Anthropometric measurement	Favorable trend
Impact	% overweight/obesity of cohort	Anthropometric measurement	Stabilization

Methodological honesty: BMI over one school year is a slow indicator: large changes are unlikely in a few months. This is why the pilot prioritizes intermediate behavioral outcomes (the gap, label reading) as an early signal, and reserves BMI for long-term follow-up. Promising drops in obesity within one school year would be dishonest.

5. Detailed Budget

Costs per school and per school year (~300 students). Figures in Mexican pesos, order of magnitude for planning.

Item	Detail	Unit Cost	Cost/School/Year
Teacher training	12 h, materials and certificate (amortized across teachers)	\$1,000/teacher	\$4,000
Student materials	Logbook + printed forms (~300 students)	\$8/student	\$2,400
Family material	9 monthly sheets (printing)	\$0.50/sheet	\$1,350
Field trips	Facilitator/chaperone and logistics (3 trips)	\$1,000/trip	\$3,000
Workshop supplies	Sugar, packaging, magnifying glasses, cardboard, etc.	—	\$1,200
Coordination & eval.	Pre/post measurement administration	—	\$2,000
TOTAL per school			~ \$13,950

Cost per student: ~ MXN\$47 per school year. Scaling: nationally (~35,000 secondary schools) the order of magnitude is ~\$490 million annually, equivalent to less than 0.1% of the federal education budget — a low-cost, high-reach intervention.

6. Risk Matrix and Mitigation

Risk	Prob.	Impact	Mitigation
Additional teacher workload generates rejection	Medium	High	Paid training + certificate with curricular credit; +1 h, not a new subject.
Low family participation	High	Medium	20-min material, not mandatory; student leads; recipes that save money.
Safety risks on field trips	Low	High	Safe radius, adult chaperone, authorized stores, conduct letter.
Effect fades over time	Medium	Medium	12-month follow-up; focus on habits, not just a campaign.
Storekeeper resistance	Low	Low	Respectful, non-accusatory approach; they are invited as informants, not blamed.
Teacher turnover between years	Medium	Medium	Self-explanatory materials and replicable training each year.

7. Governance and Sustainability

Operations define clear roles: school administration authorizes and coordinates; the Physical Education teacher implements; a student committee supports the audits; and the family accompanies. Sustainability rests on three pillars: the program is inserted into an already-existing subject (it does not depend on permanent extraordinary

budget), materials are reusable and low-cost, and the school accumulates its own audit data, generating an institutional asset that motivates continuity.

8. Pilot Measurement Instruments

To estimate the effect on the intention-action gap, the same questionnaire is applied before and after, in both intervention and control groups. These are the core items:

PRE/POST QUESTIONNAIRE – HABITS AND INTENTION (anonymous, folio: ____)

KNOWLEDGE 1. Excessive consumption of sugary drinks harms health. [Strongly agree ... Disagree] 2. Indicate how many warning seals you know: ____

INTENTION 3. How likely is it that next week you will buy a healthy option at the corner store? [Very likely ... Very unlikely]

SELF-REPORTED BEHAVIOR (last 7 days) 4. How many times did you buy a sugary drink? ____ 5. How many times did you buy fruit or water? ____ 6. Did you read the label before buying something? [Never / Sometimes / Always]

SELF-EFFICACY 7. I feel capable of choosing a healthy option even if my friends don't. [Strongly agree ... Disagree]

The intention-action gap is estimated as the difference between item 3 (intention) and reported behavior (items 4–5). Its reduction between pre and post, compared against the control group, is the key intermediate outcome of the pilot.

9. Summary Timeline (One School Year)

Month	Milestone
Jul–Aug	Phase 0: partnerships, materials, recruitment
Aug	Phase 1: teacher training and certification
Aug–Sep	Phase 2: baseline (pre-measurements)
Sep–Jun	Phase 3: 32 sessions + 3 audits + 9 family activities
Jun	Phase 4: post-measurements
Jul	Phase 5: analysis; 12-month follow-up scheduled

PART VI

Curricular Alignment

How it fits into the Nueva Escuela Mexicana without legislative reforms.

Alignment with the Nueva Escuela Mexicana (Plan 2022)

The proposal does not compete with the official curriculum: it implements it. Plan 2022 already establishes "Healthy Living" as an articulating strand, but it is rarely translated into concrete practices. This is that translation. The following matrix shows the explicit correspondence.

NEM Element	How the Program Implements It
Articulating strand: Healthy Living	Is the thread connecting all 4 blocks; operationalized in real consumption decisions.
Articulating strand: Critical Thinking	Component 2: students investigate and question their own environment with data.
Formative field: The Human and Community	The audit and family bridge connect the student to their community.
Formative field: Knowledge and Scientific Thinking	Data collection, counting, calculating proportions, and field data analysis.
Project-based methodology (NEM)	Each block is a project with an observable product; the year culminates in a portfolio.
Formative assessment	Assessed by evidence and performance, not by memorization exam.
Graduate profile (critical citizenship)	Students graduate capable of deciding autonomously and with information.

Viability argument: Proposing a new subject would require a federal curricular reform — unrealistic. Reinforcing a strand that Plan 2022 already mandates to work is immediately viable, at the school or state level, without legislative changes.

Ethics, Scaling, and Final Discussion

1. Ethical Considerations

- Consent and assent: All measurements in the pilot require informed consent from parents and assent from the minor; participation is voluntary and revocable.
- Non-stigmatization: The program never publicly singles out or weighs students; BMI is handled confidentially and in aggregate. The focus is on decisions and environment, not on bodies.
- Treatment of the storekeeper: Audits are conducted with respect and permission; the storekeeper is an informant, not a culprit. Any discourse criminalizing them is avoided.
- Data handling: All pilot data are anonymized; no individual cases are published.
- Equity: The control group is offered the program after the pilot ends (waiting list design).

2. Communication Plan

Actor	Message	Channel
Education authority	Low-cost program that implements the NEM and is measurable	Official brief and pilot results presentation
School administration	Does not add a subject; recognizes and certifies the teacher	Meeting and collaboration letter
Teachers	Paid training, ready materials, curricular certificate	Training workshop
Families	Brief invitation that saves money and cares for their children	Monthly sheets and initial meeting
Students	You will learn to decide and to research your own neighborhood	In the classroom

3. Scaling Route

The proposal is designed to grow in stages, validating before mass-scaling:

- Pilot (1 school, 1 year): Secondary School 78 with control group; validates effect on the gap and adjusts the model.
- State replication (10–20 schools): With pilot adjustments, extends in San Luis Potosí through the state educational authority.
- National scaling: Since the program is built on the NEM, it can be integrated into continuing teacher training and official materials, without legislative reform.

At the national level, the order of magnitude (~\$490 million annually for ~35,000 secondary schools) represents less than 0.1% of the federal education budget: a massive-reach, marginal-cost intervention.

4. Final Discussion

This project traveled the path from raw data to solution without hiding the missteps. A model that failed revealed that the problem was structural; machine learning exposed the intention-action paradox; our survey confirmed it; the simulation projected — with honesty about uncertainty — the impact of educating the next generation; and this manual converts that conclusion into a concrete, costed, and measurable program. The real validation will be the pilot, in a school that has already opened its doors. We do not promise to solve Mexico's obesity problem in one school year; we propose a leverage point, based on evidence, to start closing the gap between what people know and what they manage to do.

5. Declared Limitations

- Generational bridge: data are from adults; the intervention targets adolescents (reasonable assumption, not data).
- Own sample: $n=104$, regional and non-probabilistic; triangulates, does not infer nationally.
- Fixed BMI thresholds in the simulation, rather than WHO age- and sex-specific percentiles.
- Constant effect over time: the model does not model attenuation; the null scenario bounds that risk.
- External validity, not proof: using evidence from school interventions to simulate one requires pilot validation.

Annexes and Instruments

A.1 — Technical Glossary

Term	Definition
Intention-action gap	Difference between those who say they would do something (intention) and those who actually do it (action). Here: 39.1 pp.
Random Forest	ML model that combines many decision trees; its average vote reduces overfitting of a single tree.
ROC-AUC	Class separation metric: 0.5 = chance, 1.0 = perfect.
Monte Carlo	Method that repeats a random experiment thousands of times to estimate the distribution of results.
Difference-in-differences	Technique that estimates the net effect by comparing the change in the treated group against the control.
SMART goal	Specific, measurable, achievable, relevant, and time-bound goal.
BMI	Body Mass Index: weight (kg) / height squared (m ²).
KPI	Key Performance Indicator monitored to evaluate progress.

A.2 — Instruments List

- Student consumption logbook (Block 1).
- Label reading form (Block 2).
- Nanostore Audit Form (Component 2, transcribed in section 2.2).
- Field data entry and analysis template.
- Monthly family bridge sheets (9, one per month).
- Assessment rubrics by product.
- Pre/post habits and intention questionnaires (for the pilot).

A.3 — Repository and Reproducibility

All quantitative analysis from the diagnosis (Parts I and II) is open-source and reproducible. The repository includes the data, the four analysis scripts, the Monte Carlo simulation engine, the technical documentation, and the interactive dashboard.

```
mit-liftnlab-nanostores/  data/      MIT datasets + own survey + JSON parameters  src/
act1..act4 (analysis and simulation) docs/      methodology, findings, proposal, references
web/      interactive dashboard  README.md  requirements.txt  LICENSE
```

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